RDFIA 3 - a

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January 2025

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1 Introduction

This lab explores Bayesian linear regression by first presenting the theory behind such models, particularly how to derive the expression of the predictive distribution. It then delves into the role and importance of choosing the appropriate basis function in shaping the behavior of the predictive variance. Indeed, the performance of such a model heavily depends on the choice of the basis function as well as the distribution of the data points, which will be analyzed in the second part of the lab.

2 Part 1 - Linear Basis Function Model

2.1 Posterior distribution

Let n denote the number of training examples, k the number of output dimensions, and p the number of input features. We consider $X \in R^{n \times p}$ as the input matrix and $Y \in R^{n \times k}$ as the output matrix.

The *posterior* distribution of the parameters w, denoted as $p(w \mid X, Y)$, reflects our updated beliefs after observing the data X and Y. According to Bayes' rule, it is proportional to the product of the **prior distribution**, $p(w)$, which represents our initial beliefs about w and **the likelihood**, $p(Y|X, w)$, which quantifies the probability of observing the data Y given X and w. Thus, the posterior distribution is expressed as:

$$p(w \mid X, Y) \propto p(Y \mid X, w)p(w)$$

In a linear model, the relationship is given by $y_i = \Phi_i^T w + \epsilon$, where $\Phi \in R^{n \times (p+1)}$ represents the design matrix and ϵ is Gaussian noise with variance $\frac{1}{\beta} = 2\sigma^2$. This implies:

$$p(y_i \mid x_i, w) \sim \mathcal{N}(\Phi_i^T w, \beta^{-1})$$

The weight vector w is assumed to follow a Gaussian prior with variance $\alpha^{-1}I$, where α controls the spread:

$$p(w \mid \alpha) \sim \mathcal{N}(0, \alpha^{-1}I)$$

So this leads us to the posterior distribution also following a Gaussian distribution:

$$p(w \mid X, Y) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

With:

$$\boldsymbol{\Sigma}^{-1} = \alpha I + \beta \Phi^T \Phi$$

$$\boldsymbol{\mu} = \beta \boldsymbol{\Sigma} \Phi^T Y$$

When the number of data points n is zero, the posterior distribution $p(w \mid X, Y)$ reduces to the prior distribution $p(w)$. Then, as n increases, the uncertainty decreases, which is reflected in the reduction of the variance of the posterior distribution as we can see on the FIGURE 1.

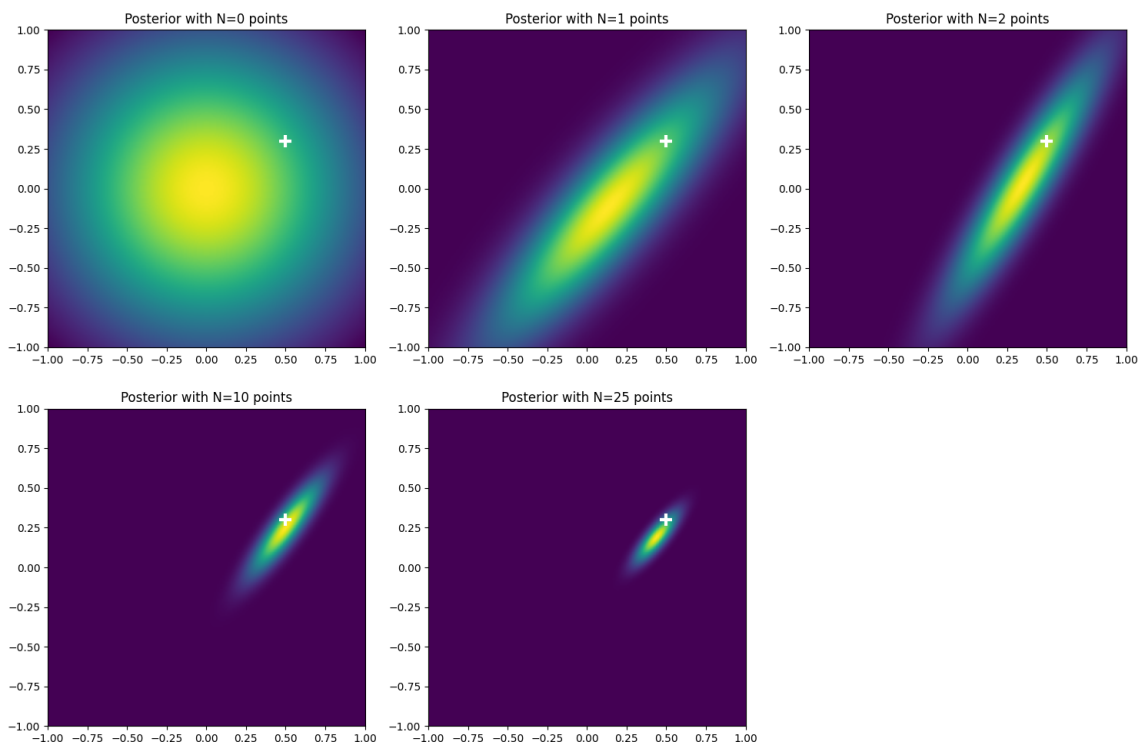


Figure 1: Posterior sampling visualization (with cross mark representing the ground truth)

2.2 Predictive distribution

In order to calculate the predictive distribution $p(y|x^*, D, \alpha, \beta)$, we use the marginalization rule that gives us:

$$p(y|x^*, D, \alpha, \beta) = \int p(y|x^*, w) p(w|D, \alpha, \beta) dw$$

Knowing that $p(w|D, \alpha, \beta) = p(w \mid X, Y) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $p(y|x^*, w) \sim \mathcal{N}(\Phi_i(x^*)^T w, \beta^{-1})$, we recognize that the predictive distribution is the convolution of two Gaussians which is also a Gaussian distribution with mean and variance given by:

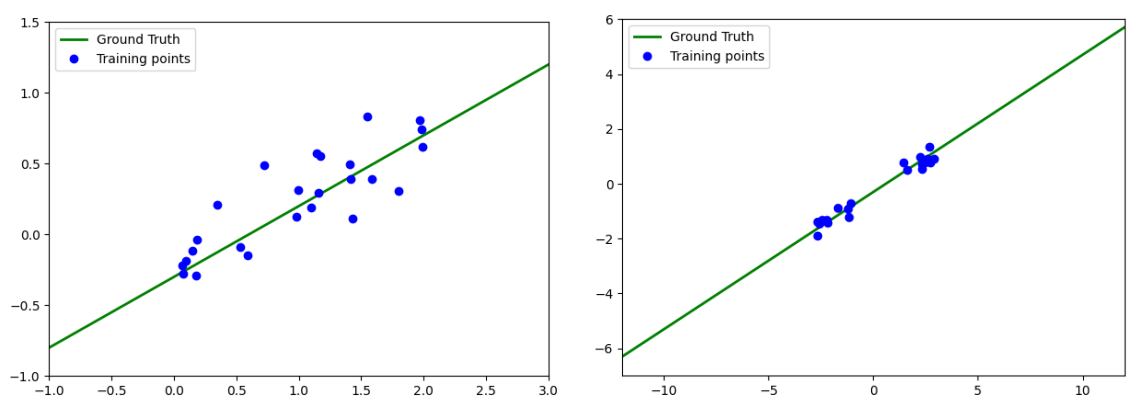
$$E[y_* \mid x_*, D, \alpha, \beta] = \Phi(x_*)^T \boldsymbol{\mu}$$

$$\mathrm{Var}[y_* \mid x_*, D, \alpha, \beta] = \beta^{-1} + \Phi(x_*)^T \boldsymbol{\Sigma} \Phi(x_*)$$

So the predictive distribution follows the Gaussian distribution:

$$y_* \mid x_*, D, \alpha, \beta \sim \mathcal{N}(\Phi(x_*)^T \boldsymbol{\mu}, \beta^{-1} + \Phi(x_*)^T \boldsymbol{\Sigma} \Phi(x_*))$$

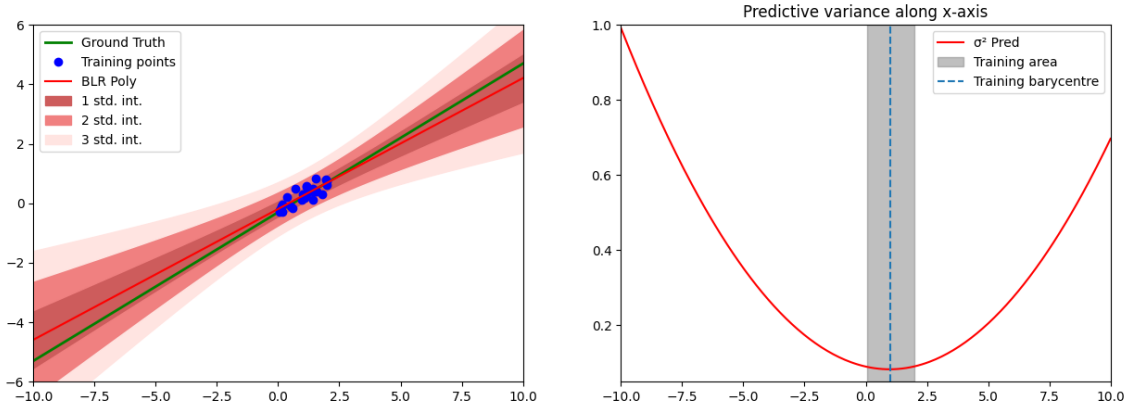
We tested this method on two different datasets represented on the FIGURE 2



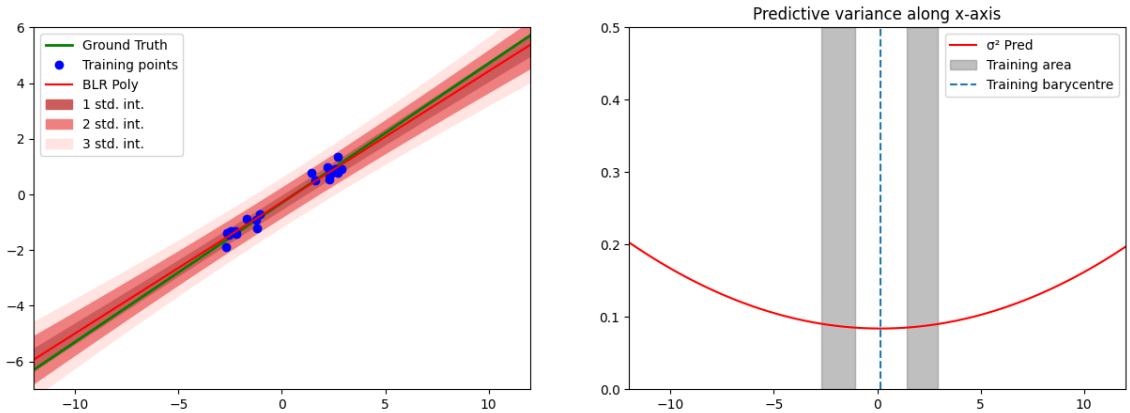
(a) (b)

Figure 2: Dataset visualization with (a) evenly distributed points and (b) points concentrated around two centers

The FIGURE 3A shows satisfying results, however we notice that the variance increases when we get far from the training distribution. This can be explained by the fact that the variance of the predictive distribution reflects the model's uncertainty about its predictions. When we move away from the region where training points are concentrated, the variance increases because the model has less information to make confident predictions. In areas with many training points, the model can better estimate the relationship between inputs and outputs, leading to lower uncertainty. However, as we move into regions with fewer or no training samples, the model **relies more on the prior distribution**, which introduces greater uncertainty.



(a) Predictions with dataset 2-(a)



(b) Predictions with dataset 2-(b)

Figure 3: Predictive distribution

We demonstrate it analytically with $\alpha = 0$ and $\beta = 1$:

$$
\begin{aligned}
\boldsymbol{\Sigma}^{-1} &= \Phi^T \Phi \\
&= \begin{pmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \\
&= \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}
\end{aligned}
$$

We invert this matrix leading us to :

$$\boldsymbol{\Sigma} = \frac{1}{\det \boldsymbol{\Sigma}^{-1}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}$$

Finally,

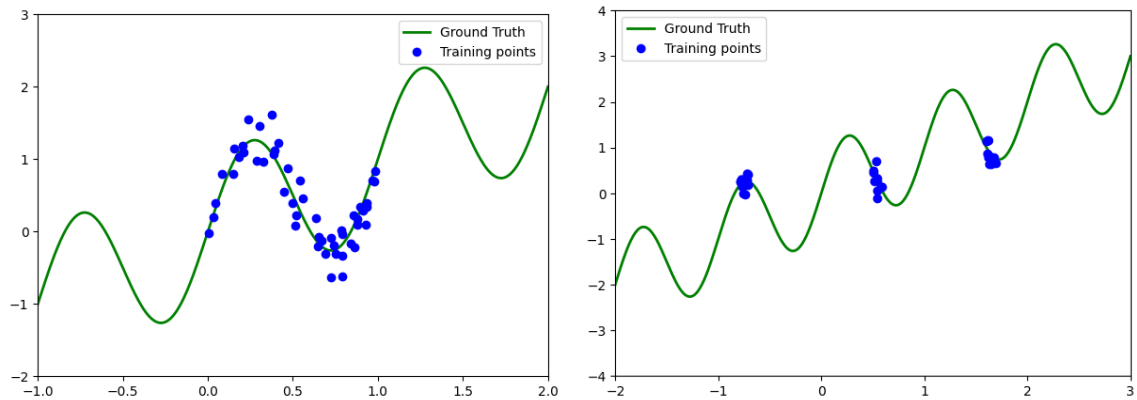
$$\begin{aligned}
\sigma^2 = \Phi(x^\star)^T \boldsymbol{\Sigma} \Phi(x^\star) &= \frac{1}{\det \boldsymbol{\Sigma}^{-1}} \begin{pmatrix} 1 \\ x^\star \end{pmatrix} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix} \begin{pmatrix} 1 & x^\star \end{pmatrix} \\
&= \frac{1}{\det \boldsymbol{\Sigma}^{-1}} \left(\sum_{i=1}^n x_i^2 - x^\star \sum_{i=1}^n x_i + x^{\star 2} n - x^\star \sum_{i=1}^n x_i \right) \\
&= \frac{1}{\det \boldsymbol{\Sigma}^{-1}} \left(\sum_{i=1}^n x_i^2 - 2x^\star \sum_{i=1}^n x_i + x^{\star 2} n \right) \\
&= \frac{1}{\det \boldsymbol{\Sigma}^{-1}} \sum_{i=1}^n \left(x_i - x^\star \right)^2
\end{aligned}$$

As we can see on the expression of the variance obtained, when x^* moves away from the training area, $(x_i - x)^2$ increases leading to an increase of the predictive variance.

The dataset 2b contains data points centered in two clusters, and the results presented in FIGURE 3B show that the predictive variance increases less when we get away from the training points than when we used the dataset 2a. This overconfidence might come from the model's strong prior belief in the data's linearity, as a line passing through distant points provides more certainty about its direction than one passing through closely spaced points.

3 Part 2 - Non Linear Models

In this part we use a different, more complex dataset represented on the FIGURE 4A (an increasing sinusoid) that we derive into the dataset 4b in order to study the influence of more dispersed data points on the predictive variance.



(a) (b)

Figure 4: Dataset visualization with (a) evenly distributed points and (b) points concentrated around three centers

3.1 Polynomial basis function

Since the closed-form expressions for the posterior and predictive distribution in the polynomial basis case resemble those of the linear model, similar conclusions can be drawn: uncertainty increases as we move away from the training data. However, the FIGURE 5 shows that the use of a polynomial kernel results in higher predictive variance, reflecting greater epistemic uncertainty. On the positive side, this basis function allows the model to capture more complex patterns, such as sinusoidal variations. Nevertheless, the model struggles to generalize beyond the training data, as it lacks sufficient information to accurately represent the periodic nature of the true function.

As we can see on the FIGURE 6, using more dispersed data points doesn't change the fact that the variance is minimal on these clusters of data. However the predictions lack a lot of accuracy outside of these clusters: the model is not able to capture

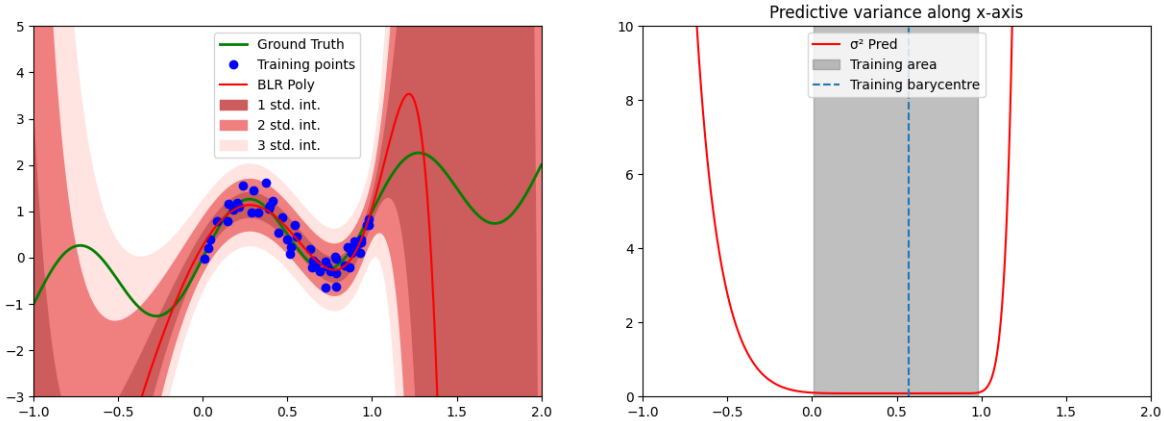


Figure 5: Predictive distribution on dataset 4a with polynomial basis function

the characteristics of the data since there are not enough points to follow efficiently the curve of the sinusoid.

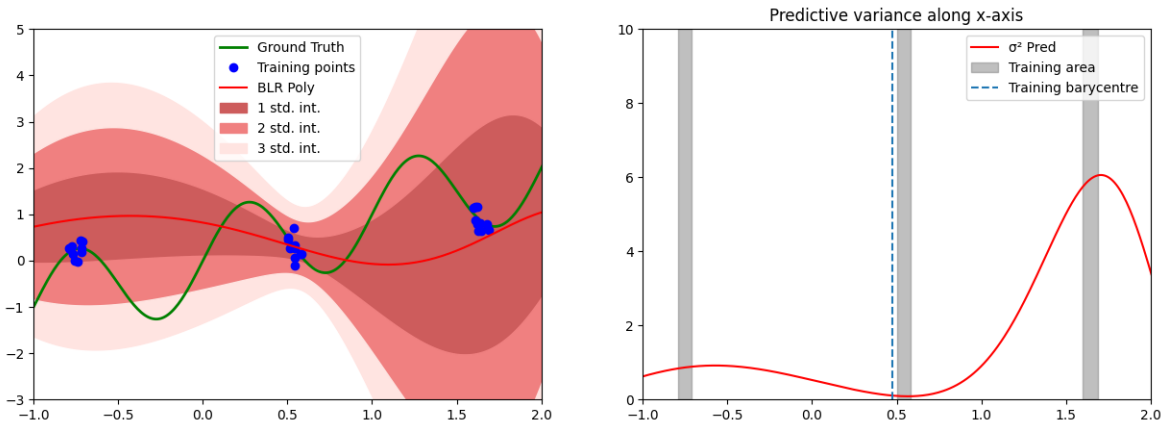


Figure 6: Predictive distribution on dataset 4b with polynomial basis function

3.2 Gaussian basis function

As we can see on the FIGURE 7, unlike previous models, the predictive variance is not highest at the edges of the data range but is instead concentrated within the region covered by the training data. The predictive variance fluctuates, with peaks and troughs reflecting changes in the density of training points. As a result, the

model tends to align closely with the mean of the training data, which influences its ability to capture sinusoidal patterns.

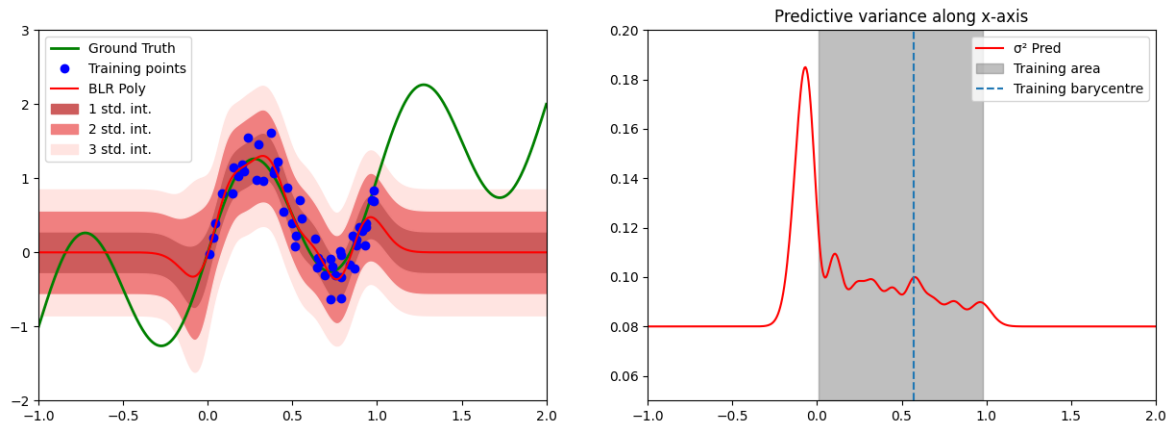


Figure 7: Predictive distribution on dataset 4a with Gaussian basis function

Visualizing the model with fewer data points highlights the previously discussed observations. Moreover, the model finds it challenging to capture the influence of extreme data points, as they fall outside the predefined mean range. However, as expected, the predictive variance increases significantly in regions without data points.

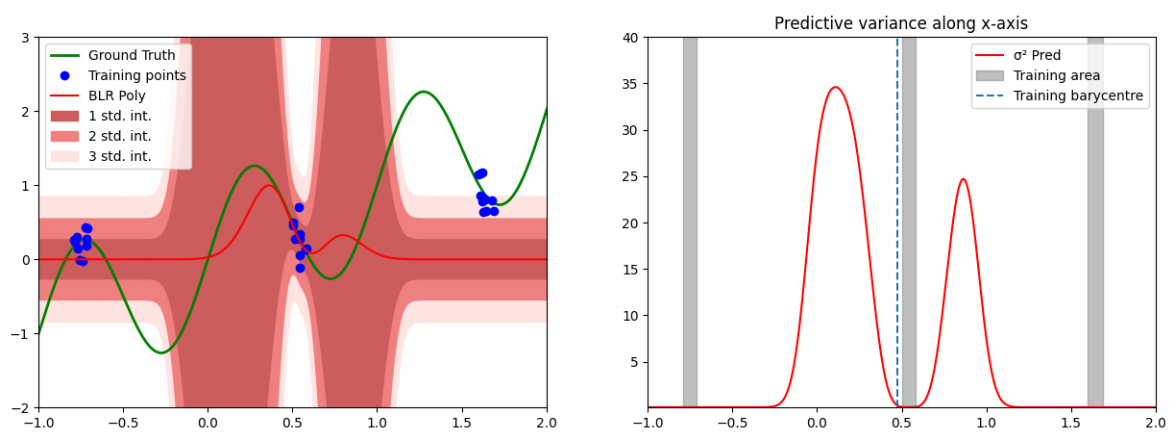


Figure 8: Predictive distribution on dataset 4b with Gaussian basis function

Far from the training distribution, the predictive variance converges to the value 0.08 as we can see on the FIGURE 7. Indeed, if we take localized basis function such as Gaussian $\left(\Phi_j(x_i) = \exp\left(-\frac{(x_i-\mu_j)^2}{2s^2}\right)\right)$, we notice that the epistemic uncertainty

$\Phi(x^*)^T \Sigma \Phi(x^*)$ converges to 0 if x^* gets far from μ_j. As a consequence, the expression of the predictive variance boils down to β^{-1} which is equal to $2\sigma^2 = 0.08$ pour $\sigma = 0.2$.

The choice of hyperparameters μ and s is crucial for effective modeling. By adjusting the placement of μ within an appropriate range, the model achieves improved results, outperforming polynomial regression while maintaining interpretability.